RAJ PATEL

raj.patel.connects@gmail.com | linkedin.com/in/raj-patel5 | Ahmedabad, Gujarat, India
portfolio-raj-25.netlify.app | github.com/R204570 | kaggle.com/raj072004

SUMMARY

AI Engineer with a B.E. in Information Technology (2026) and a Minor in Data Science, building agentic LLM and RAG pipelines, computer vision models and production web applications that are deployed rather than running on localhost. Delivered a YOLOv8 PCB inspection system (mAP@0.5 of about 0.90) during an Intel internship, a 3D Attention U-Net and RAG clinical decision support system, an open-source MCP documentation server (DocsForge) and a Django hospital management system with 100+ GitHub stars. Seeking AI Engineer, Agentic AI and ML roles.

EDUCATION

Bachelor of Engineering in Information Technology — Graduated 2026

SAL Engineering and Technical Institute, Gujarat Technological University (GTU), Ahmedabad — *CGPA: 8.77*

Minor/Honours in Data Science (GTU advanced track), Qualified — Graduated 2026

TECHNICAL SKILLS

Programming	Python (pandas, NumPy, Matplotlib), JavaScript, SQL, CSS, Tailwind CSS
Machine Learning	Machine Learning, Deep Learning, Computer Vision, Generative AI, Agentic AI, Retrieval-Augmented Generation (RAG), PyTorch, TensorFlow, Keras, scikit-learn, YOLOv8, 3D Attention U-Net, Hugging Face, sentence-transformers, feature engineering, model deployment
LLM and Agent Tooling	Model Context Protocol (MCP), LLM APIs (Claude, OpenAI, Gemini, Groq), Ollama, prompt engineering, vector search (pgvector), Claude Code
Backend and Web	Django, Flask, FastAPI, Streamlit, Node.js, AngularJS, REST APIs
Databases	SQL: PostgreSQL, MySQL, SQLite, BigQuery; NoSQL: MongoDB
Cloud and DevOps	Google Cloud Platform, Vercel, Netlify, Render, Git, GitHub, GitHub Actions (CI)
Tools and Testing	Linux (command line), Jupyter, Google Colab, Kaggle, VS Code, pytest, unittest

EXPERIENCE

AI for Manufacturing Intern (Certificate) — Feb 2025 – Jul 2025

Intel Technology India Pvt. Ltd. with GTU and GCCI — *Online, 16 weeks*

- Completed a 16-week program, part of the first batch of Intel's university internship and upskilling initiative, applying AI to manufacturing quality problems.
- Built a PCB defect inspection system with YOLOv8 for smartphone assembly, reaching mAP@0.5 of about 0.90 across 10 defect classes with multi-dataset training, and optimized inference for production settings.
- Delivered a Streamlit application for real-time webcam and batch inspection; invited to present the project to Dr. Rajul K. Gajjar (Vice Chancellor, GTU), Anshul Sonak and Shweta Khurana (Intel).

AI/ML Intern (Certificate) — Jan 2026 – Apr 2026

Microsoft Elevate with GTU, via FICE Education Pvt. Ltd. — *Online, 12 weeks*

- Completed the 12-week Microsoft Elevate and GTU internship (21 Jan to 10 Apr 2026) on artificial intelligence and machine learning, Microsoft Copilot and other emerging technologies.
- Built the NeuroAssist capstone end to end (3D Attention U-Net segmentation, RAG-grounded LLM reporting, Flask and PostgreSQL web application) and documented it in a 98-page report with UML class and data flow diagrams.

CERTIFICATIONS AND ACHIEVEMENTS

- Invited to present the Intel and GTU AI for Manufacturing project to Dr. Rajul K. Gajjar (Vice Chancellor, Gujarat Technological University), Anshul Sonak (Head and Global Director, Intel Digital Readiness Programs, Intel Corporation) and Shweta Khurana (Sr. Director, Asia Pacific and Japan, Government Partnerships and Initiatives, International Government Affairs Group, Intel). (LinkedIn)
- Anthropic Academy: Claude 101, Claude Code 101, Introduction to MCP. (Certificates)
- Google AI Professional Certificate; Google Cloud Skills Boost badges. (Certificate | Badges)
- IELTS: overall band 7.0 (CEFR C1). (Score report)

PROJECTS

NeuroAssist: Brain Tumor Detection and Clinical Decision Support (GitHub) Mar 2026 – Apr 2026

- Trained a 3D Attention U-Net on BraTS 2020 (T1, T1ce, T2, FLAIR) to segment tumor sub-regions; derived tumor volumes, diameter, centroid and lobe involvement from the predicted masks.
- Built a RAG pipeline (sentence-transformers embeddings and pgvector over curated WHO 2021, VASARI and RANO references) with a Groq-hosted LLM that turns a structured neuro-oncology intake and imaging metrics into a WHO-aligned diagnostic report.
- Developed a Flask and PostgreSQL web application with NIfTI and DICOM upload, intake-completeness scoring and an annotated-slice report viewer; added unit tests for LLM response parsing and MRI modality detection.

Tech: Python, TensorFlow/Keras, Flask, PostgreSQL, pgvector, sentence-transformers, Groq API, NIfTI/DICOM

DocsForge: Open-Source MCP Server for LLM-Ready Documentation (Website | GitHub) 2026 – Present

- Built an MCP server, with a CLI and web chat sharing one extraction engine, that finds a library's official documentation from its name via npm, PyPI and crates.io, verifies the page documents that package, and harvests version-specific Markdown for Claude Code, Cursor and other AI agents.
- Evaluated rather than demoed: 91% of 56 unseen library names resolved to the correct documentation on the first try, backed by a 53-site field test across documentation generators, a 76-case live MCP benchmark and 1,310 offline tests.
- Hardened for hosted use: SSRF checks on every redirect hop, bearer-token and OAuth 2.1 authentication, six LLM provider integrations and Vercel serverless deployment; MIT-licensed.

Tech: Python, FastAPI, MCP, PostgreSQL, Playwright, GitHub Actions, pytest, Vercel

Optical PCB Defect Inspection for Smartphone Assembly (Live Demo | GitHub) May 2025 – Jun 2025

- Co-built a YOLOv8 defect detector with mAP@0.5 of about 0.90 across 10 defect classes during the Intel AI for Manufacturing program, using a multi-dataset training workflow.
- Deployed a Streamlit application (Python, OpenCV, Plotly) for single-image, batch and live-camera inspection, with defect analytics charts and ZIP export of annotated images.

Tech: Python, YOLOv8 (Ultralytics), PyTorch, OpenCV, Streamlit, Plotly, pandas

Hospital Management System (SmartCare HMS) (Live Demo | GitHub) Mar 2025 – Aug 2026

- Built an open-source Django 4.2 system (MIT, 100+ GitHub stars) for a multi-specialty hospital covering patient records, appointments with doctor availability and leave approval, admissions with room and bed management, billing, and a public website.
- Enforced role-based access for five staff roles (administrator, doctor, nurse, receptionist, pharmacist) with custom middleware and per-view decorators; services and selectors layers; dev and prod settings.
- Implemented an in-house pharmacy: nurse-to-pharmacy drug requests (pending, approved, dispensed) with atomic stock deduction, retail invoices, purchase orders with batch and expiry tracking, PDF reports and an admin analytics dashboard. Deployed on Render.

Tech: Python, Django, Bootstrap 5, SQLite, Channels/Daphne (ASGI), xhtml2pdf, SMTP/IMAP

Additional Projects

- **Cafe POS** (GitHub): cafe point-of-sale with QR-code table ordering. Django, PostgreSQL, DRF.
- **Reconciliation** (GitHub): wallet reconciliation with overdraft checks, audit trail and dashboard. Flask, SQL Server.
- **SAC AI Search** (GitHub): hybrid TF-IDF retrieval and LLM agent mapping service descriptions to GST codes. Python, scikit-learn, PostgreSQL.
- **ExpenseFlow** (Live Demo | GitHub): expense tracker with Groq-powered receipt processing and bill splitting. React, Node.js, Express, Prisma.
- **GearGuard** (GitHub): equipment and maintenance management platform. Django 5.2, PostgreSQL.
- **BraTS 2020 MGMT Predictor** (Kaggle Dataset | Training Notebook): MGMT-methylation classifier (custom 3D ResNet, PyTorch).